

VTOL tail-sitter V2

NY

Summer 2026

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Nomenclature**Assumptions****Constraints****Cost:**

The total component budget for this build is to fit under \$400.

Dimensions:

- Wingspan of 54 inches/
- Maximum weight of 1.3kg, driven by hardware motor constrain
- An aspect ratio of 14 is chosen to authentically represent Iteration 1's design

Hardware:

- Main motor: KDE3510XF-475 (Max thrust of 22 N)

Initial Sizing and aircraft specification calculations

Assuming a reasonably efficient wing planform ($e \approx 0.8$), and an estimated initial zero-lift drag coefficient of $C_{d_0} = 0.03$, we have the folling basic aircraft metrics:

$$b = 1.3761 \text{ m}$$

$$AR = 14$$

$$S = 0.135 \text{ m}^2$$

$$W = 12.955 \text{ N}$$

$$\rho = 1.225 \text{ kg/m}^3$$

$$k = \frac{1}{\pi A Re} \approx 0.02842$$

Cruise

To maximize endurance, the aircraft will fly at the minimum-drag velocity:

$$v_{md} = \sqrt{\frac{2n_z W}{\rho S C_{L_{md}}}}$$

$$n_z = 1$$

$$C_{L_{md}} = \sqrt{\frac{C_{d_0}}{k}} \approx 1.0274$$

$$\boxed{v_{md} = 12.188 \text{ m/s}}$$

Thrust requirements, therefore follow as:

$$T = \frac{1}{2} \rho v^2 S C_D$$

$$C_D = C_{d_0} + k C_{L_{md}}^2 \approx 0.06$$

$$\boxed{T_{req} = 0.757 \text{ N}}$$